ASSEMBLY OF THE LOCAL TYPE II EXCLUSION THEOREM FOR NAVIER–STOKES BLOWUP

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Abstract. We present an explicit assembly of a local Type II exclusion argument for the three-dimensional incompressible Navier–Stokes equations in PDE form.

From a suitable weak solution near a possible singular point we obtain positive local Caffarelli–Kohn–Nirenberg concentration. Assuming the local Type I tangent criterion, we split off the Type I tangent alternative and assemble the remaining local Type II branch exclusions proved in the preceding papers.

The alternatives are:

(1) compact single-core with finite local cost;
(2) multibubble/cascade or gauge-degenerate splitting;
(3) rough-core loss of local H^1 control;
(4) scale collapse.

Each branch is routed through explicit local lemmas and through the branch-closure results established earlier in the series.

1. Introduction

Let (u, p) be a suitable weak solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

in a neighborhood of $z_0 = (x_0, T)$. Assume

$$u \in L^\infty_{\mathrm{loc}} L^2_{\mathrm{loc}}, \quad \nabla u \in L^2_{\mathrm{loc}}, \quad p \in L^{3/2}_{\mathrm{loc}},$$

and the local energy inequality.

For $r > 0$, set

$$\begin{aligned} Q_r(z_0) &:= B_r(x_0) \times (T - r^2, T), \\ C(z_0, r) &:= r^{-2} \int_{Q_r(z_0)} |u|^3, \\ D(z_0, r) &:= r^{-2} \int_{T-r^2}^{T} \int_{B_r(x_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2}. \end{aligned}$$

Define $\mathcal{M}(z_0, r) := C(z_0, r) + D(z_0, r)$.

Lemma 1.1 (Local CKN ε-regularity)**.** *There is $\varepsilon_{\mathrm{CKN}} > 0$ such that $\mathcal{M}(z_0, r) < \varepsilon_{\mathrm{CKN}}$ implies $u \in L^\infty(Q_{r/2}(z_0))$; see* [CKN82, Lin98].

2. Standing Type I input

Assumption 2.1 (Local Type I tangent criterion)**.** At a singular point with positive local CKN concentration, every nonzero bounded parabolic tangent limit belongs to the local Type I class and is excluded by the Type I part of the theory.

All other inputs used below are stated as lemmas or theorems in the preceding Type II papers: the repaired-gauge representation, compact-ball pressure decomposition, single-core exclusion, terminal active-frame decoupling, rough-core Caccioppoli regularity, and the scale-collapse stratification.

3. Auxiliary quantitative and structural lemmas

Given suitable (v, q) on $Q_R = B_R \times (-R^2, 0)$, define

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v|^2, \quad E_v(R) = R^{-1} \int_{-R^2}^{0} \int_{B_R} |\nabla v|^2,$$

$$C_{v,q}(R) = R^{-2} \int_{Q_R} |v|^3, \quad D_{v,q}(R) = R^{-2} \int_{Q_R} |q - (q)_{B_R}(s)|^{3/2}.$$

Lemma 3.1 (Local energy with cutoff)**.** *Let $\phi \in C_c^\infty(Q_R)$, $\phi \geq 0$, and (v, q) suitable on Q_R. For $-R^2 < \tau_1 < \tau_2 < 0$,*

$$\begin{aligned}\frac{1}{2} \int_{B_R} |v(y, \tau_2)|^2 \phi^2 - \frac{1}{2} \int_{B_R} |v(y, \tau_1)|^2 \phi^2 + \nu \int_{\tau_1}^{\tau_2} \int_{B_R} |\nabla(v\phi)|^2 \\ \leq \int_{\tau_1}^{\tau_2} \int_{B_R} \Big(|v|^3 (|\nabla \phi| + |\nabla \phi|^2) + |q - (q)_{B_R}|\, |v|\, |\nabla \phi| \Big) \\ + \int_{\tau_1}^{\tau_2} \int_{B_R} |v|^2 \phi \left(|\partial_\tau \phi| + |\Delta \phi| \right).\end{aligned}$$

Proof. Test the local equation with $v\phi^2$, use $\nabla \cdot v = 0$, and integrate by parts in space-time over $Q_R \cap (\mathbb{R}^3 \times (\tau_1, \tau_2))$. This is the standard localization identity for suitable weak solutions. □

Lemma 3.2 (Pressure decomposition on a ball)**.** *Let (v, q) be suitable on B_R. Decompose, after subtracting a function of time,*

$$q = q_{\mathrm{loc}} + q_{\mathrm{har}},$$

where

$$-\Delta q_{\mathrm{loc}} = \partial_i \partial_j (v_i v_j) \quad \text{in } B_R, \qquad q_{\mathrm{loc}}\big|_{\partial B_R} = 0, \qquad \Delta q_{\mathrm{har}} = 0 \quad \text{in } B_R.$$

Then

$$\|q_{\mathrm{loc}} - (q_{\mathrm{loc}})_{B_R}\|_{L^{3/2}(B_R)} \leq C \|v\|_{L^3(B_R)}^2,$$

and, for every $0 < \theta < 1$,

$$\|\nabla q_{\mathrm{har}}\|_{L^{3/2}(B_{\theta R})} \leq C_\theta R^{-1} \|q_{\mathrm{har}} - (q_{\mathrm{har}})_{B_R}\|_{L^{3/2}(B_R)}.$$

Proof. The local part follows from the Calderón–Zygmund estimate for the Dirichlet Poisson problem. The harmonic part follows from the standard interior gradient estimate for harmonic functions. □

Lemma 3.3 (Caccioppoli control of energy from concentration)**.** *For any $0 < \theta < 1$, there exists $C > 0$ such that*

$$A_v(\theta R) + E_v(\theta R) \leq C\Big(1 + A_v(R) + E_v(R) + C_{v,q}(R) + D_{v,q}(R)\Big).$$

Equivalently, a bounded scale-invariant CKN functional on Q_R gives bounded local energy on $Q_{\theta R}$.

Proof. Choose $\phi \in C_c^\infty(B_R)$ with $\phi \equiv 1$ on $B_{\theta R}$, $|\nabla \phi| \lesssim R^{-1}$, $|\Delta \phi| \lesssim R^{-2}$. Apply Theorem 3.1 on $(-R^2, 0)$, bound the convection and pressure terms via Hölder–Young, and absorb a small portion of $\|\nabla(v\phi)\|_{L^2}$ into the left-hand side. □

Lemma 3.4 (Compactness from scale-invariant local bounds)**.** *Let (v_n, q_n) be suitable on Q_R and satisfy, for some constants C_0, C_1, C_2,*

$$\sup_n A_{v_n}(R) + \sup_n E_{v_n}(R) < \infty, \quad \sup_n C_{v_n,q_n}(R) < \infty, \quad \sup_n D_{v_n,q_n}(R) < \infty,$$

$$\sup_n \|q_n\|_{L_t^{3/2} L_x^{3/2}(Q_R)} < \infty, \quad \sup_n \|\partial_s v_n\|_{L_t^{3/2} H_x^{-1}(Q_R)} < \infty,$$

where $\partial_s v_n$ is computed from the renormalized PDE and pressure decomposition $q_n = q_n^{\mathrm{loc}} + q_n^{\mathrm{har}}$. Then a subsequence satisfies

$$v_n \to v \text{ in } L^2_{\mathrm{loc}}(Q_R), \quad v_n \rightharpoonup v \text{ in } L_t^2 H_x^1(Q_R), \quad q_n \rightharpoonup q \text{ in } L_t^{3/2} L_x^{3/2}(Q_R).$$

Proof. Bounds on A and E place v_n in $L_t^\infty L_x^2 \cap L_t^2 H_x^1$. The equation with bounded drift/convection terms and $q_n \in L_t^{3/2} L_x^{3/2}$ gives a uniform bound on $\partial_s v_n$ in $L_t^{3/2} H_x^{-1}$ after pressure splitting by Theorem 3.2. Aubin–Lions–Simon compactness yields the strong and weak convergences [Sim87]. □

Lemma 3.5 (Zero dissipation)**.** *If $W \in L_t^2 H_x^1(Q_R)$ and $\int_{Q_R} |\nabla W|^2 = 0$, then for a.e. τ, $W(\cdot, \tau)$ is spatially constant on B_R.*

Proof. $\nabla W = 0$ almost everywhere, hence every spatial slice is a.e. constant. □

Lemma 3.6 (Separated profile decoupling)**.** *Let (f_n) and (g_n) be two families in $L^3(Q_R)$. Assume either* $\mathrm{supp} f_n$ *and* $\mathrm{supp} g_n$ *are separated at fixed scale, or one is observed in a frame that is scale-separated from the other. Then*

$$\int_{Q_R} |f_n + g_n|^3 = \int_{Q_R} |f_n|^3 + \int_{Q_R} |g_n|^3 + o(1),$$

and similarly for local pressure norms in $L^{3/2}$.

Proof. The elementary inequalities

$$\big||a+b|^3 - |a|^3 - |b|^3\big| \le C\big(|a|^2|b| + |a||b|^2\big)$$

reduce the claim to the mixed integrals. These mixed terms are controlled by Hölder's inequality and by the overlap of the two supports after placing both profiles in the same scale. The overlap tends to zero under the stated separation hypothesis. The pressure assertion follows from the same argument after applying Theorem 3.2 to the local pressure components. □

4. Scale-collapse auxiliary setup

For a renormalized branch (V, P, a, b, λ) set

$$a_+(\tau) := \max\{a(\tau), 0\}, \qquad a_-(\tau) := \max\{-a(\tau), 0\},$$

$$C_{\mathrm{sc}}(\tau) := a_-(\tau) M_R(\tau), \quad C_{\mathrm{abs}}(\tau) := \nu \int_{B_R} |\nabla V|^2 \, dy + |a(\tau)| M_R(\tau),$$

where $M_R(\tau) := \int_{B_R} |V(y,\tau)|^2 \, dy$.

Definition 4.1 (Scale-collapse and finite-cost)**.** A branch has finite scale-collapse cost if

$$\int_{\tau_0}^\infty C_{\mathrm{sc}}(\tau) \, d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^\infty C_{\mathrm{abs}}(\tau) \, d\tau < \infty.$$

Lemma 4.2 (Collapse with nonvanishing localized core). *Assume $\lambda(\tau) \to 0$ and there exists $m_0 > 0$, $\tau_1 \geq \tau_0$ with $M_R(\tau) \geq m_0$ for a.e. $\tau \geq \tau_1$. Then*

$$\int_{\tau_0}^{\infty} C_{\mathrm{sc}}(\tau)\, d\tau = \infty.$$

Proof. By $\frac{d}{d\tau} \log \lambda = a$, we have $\int_{\tau_1}^{\tau_2} a \to -\infty$ as $\tau_2 \to \infty$. Since $a \geq -a_-$, this forces $\int_{\tau_1}^{\infty} a_-(\tau)\, d\tau = \infty$. Multiply by $M_R \geq m_0$. □

Lemma 4.3 (Absolute-cost domination). *For almost every τ, $C_{\mathrm{sc}}(\tau) \leq C_{\mathrm{abs}}(\tau)$. Hence infinite scale-collapse cost implies infinite absolute cost.*

Proof. $C_{\mathrm{abs}} = \nu \int |\nabla V|^2 + |a| M_R = a_+ M_R + a_- M_R + \nu \int |\nabla V|^2 \geq a_- M_R = C_{\mathrm{sc}}$. □

Lemma 4.4 (Scale-collapse stratification input). *A repaired-gauge scale-collapse branch with a nonvanishing localized core enters the scale-collapse stratum of the preceding analysis. In that stratum the divergent quantity $C_{\mathrm{sc}} = a_- M_R$ is not treated as a perturbative error: it is the scale-drift alternative itself. The remaining alternatives are the thin-drift, oscillatory-modulation, loss-of-compactness, and scale-rigidity cases isolated by the local stratification.*

Proof. This is the local scale-collapse stratification established in the preceding analysis: genuine collapse with a retained localized core forces divergence of the negative scale-drift functional, and the stratification treats this divergence as an alternative rather than as a finite error term. □

5. Positive local concentration and Type II sequence

Theorem 5.1 (Positive local concentration at singular points). *If z_0 is singular, then*

$$\limsup_{r \downarrow 0} \mathcal{M}(z_0, r) > 0.$$

Equivalent to existence of $\eta > 0$ and $r_n \downarrow 0$ with $\mathcal{M}(z_0, r_n) \geq \eta$.

Proof. Assume $\limsup = 0$. Then some $r_0 > 0$ has $\mathcal{M}(z_0, r) < \varepsilon_{\mathrm{CKN}}$ for $0 < r < r_0$. By Theorem 1.1, every $Q_{r/2}(z_0)$ is regular, contradiction. □

Set

$$u_n(y, s) := r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y, s) := r_n^2 p(x_0 + r_n y, T + r_n^2 s),$$

for a sequence $r_n \downarrow 0$ with $\mathcal{M}(z_0, r_n) \geq \eta$. Then (u_n, p_n) is suitable on every fixed backward cylinder.

Definition 5.2 (Local Type II sequence). The sequence (r_n) is a local Type II sequence if $\mathcal{M}(z_0, r_n) \geq \eta > 0$ and no localized window with uniform bounds on compact parabolic cylinders has a nonzero bounded parabolic tangent limit.

6. Concentration-compactness dichotomy

Theorem 6.1 (Local concentration-compactness dichotomy). *After a subsequence of (u_n, p_n), either*

(i) *(uniform scale-invariant local bounds) For every $R < \infty$,*

$$\sup_n \big(A_{u_n}(R) + E_{u_n}(R) + C_{u_n, p_n}(R) + D_{u_n, p_n}(R) \big) < \infty;$$

(ii) *(multibubble/cascade) there exist $R_0 < \infty$, $\eta > 0$, points $(y_n, s_n) \in Q_{R_0}$, and radii $\rho_n \downarrow 0$ such that*

$$\rho_n^{-2} \int_{Q_{\rho_n}(y_n, s_n)} \big(|u_n|^3 + |p_n - (p_n)_{B_{\rho_n}(y_n)}|^{3/2} \big)\, dy\, ds \geq \eta.$$

Proof. Set $\mathcal{B}_n(R) := A_{u_n}(R) + E_{u_n}(R) + C_{u_n,p_n}(R) + D_{u_n,p_n}(R)$.

Case (i). Suppose $\sup_n \mathcal{B}_n(R) < \infty$ for every R. From the renormalized equation and Theorem 3.2, we have a uniform bound on $\partial_s u_n$ in $L_t^{3/2} H_x^{-1}$ on each compact parabolic cylinder. Then Theorem 3.4 applies; diagonalizing over R gives compactness on every compact parabolic cylinder, so item (i) holds.

Case (ii). Assume there is R_0 with $\sup_n \mathcal{B}_n(R_0) = \infty$. If both $C_{u_n,p_n}(R_0)$ and $D_{u_n,p_n}(R_0)$ were bounded, then $A_{u_n}(R_0) + E_{u_n}(R_0)$ would still yield the required scale-invariant local bounds by Theorem 3.3, contradiction. Hence after extracting one time-scale and center of concentration and a scale $\rho_n \downarrow 0$, $\rho_n^{-2} \int_{Q_{\rho_n}} |u_n|^3$ or the pressure term is bounded below by $\eta > 0$, which yields item (ii) by definition of localized CKN mass. □

7. Local PDE components

Theorem 7.1 (Local single-core criterion)**.** *A local Type II sequence cannot satisfy simultaneously:*

(i) *uniform scale-invariant bounds on compact parabolic cylinders in Q_{R_*};*
(ii) *nondegenerate local scale–translation gauges on compact time intervals;*
(iii) *pressure reconstruction modulo time functions;*
(iv) *local Caccioppoli structure;*
(v) *finite localized Type II cost.*

Proof. Let renormalized (V_n, P_n) be defined on $\mathcal{I}_n \times B_{R_*}$ with coefficients a_n, b_n, and local windows bounded in $A + E + C + D$.

Step 1: compact extraction. By uniform boundedness of $A + E + C + D$ and bounded $\partial_\tau V_n$ from the equation, Theorem 3.4 gives, after subsequence, $(V_n, P_n) \to (\bar{V}, \bar{P})$ locally with $V_n \to \bar{V}$ in L^2_{loc}, $V_n \rightharpoonup \bar{V}$ in $L_t^2 H_x^1$.

Step 2: zero local dissipation. Finite localized Type II cost gives the vanishing local Caccioppoli cost along this branch, so by weak lower semicontinuity $\int |\nabla \bar{V}|^2 = 0$. Hence by Theorem 3.5, $\bar{V}(\tau, \cdot)$ is spatially constant for a.e. τ.

Step 3: eliminate both constants. If $\bar{V} \equiv 0$, then by strong L^2-convergence local CKN mass drops: $C_{V_n,P_n}(1) + D_{V_n,P_n}(1) \to 0$, contradicting the positive local concentration inherited from u_n.

If $\bar{V} \not\equiv 0$, undoing the renormalization gives a nonzero bounded parabolic tangent profile, contradicting Theorem 5.2. Both options are impossible. □

Theorem 7.2 (Localized scale–translation representation)**.** *On a nondegenerate single-core Type II sequence there exists smooth $\lambda(\tau) > 0$, $x_c(\tau)$, and local time map $dt = \lambda^2 d\tau$ such that*

$$V(y, \tau) = \lambda u(x_c + \lambda y, t), \quad P(y, \tau) = \lambda^2 p(x_c + \lambda y, t) + \pi(\tau),$$

solve

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(V + y \cdot \nabla V) + b \cdot \nabla V, \quad a = \lambda'/\lambda,\ b = -x_c'/\lambda.$$

If the Jacobian degenerates, one is in the gauge-degenerate multibubble alternative.

Proof. Set $x = x_c(\tau) + \lambda(\tau) y$. Then $\nabla_x = \lambda^{-1} \nabla_y$ and $\partial_t = \lambda^{-2} \partial_\tau - \lambda^{-1} a(y \cdot \nabla) + \lambda^{-1} b \cdot \nabla$. Direct substitution into NSE gives the equation above. □

Theorem 7.3 (Local multibubble and cascade exclusion)**.** *No local multibubble, cascade, or gauge-degenerate Type II configuration persists after the single-core and decoupling reductions.*

Proof. Assume such a configuration. By Theorem 6.1, either compactness fails and we are already in an active-core splitting regime, or case (i) holds and a single core is selected.

Active splitting. Apply Theorem 3.6 on the selected window decomposition $(u_n, p_n) = \sum_{j=1}^{J} U_n^{(j)} + R_n$. For each $j \neq k$, cross terms vanish in the localized L^3- and $L^{3/2}$-norms. Thus positive local mass either concentrates in one dominant component or in separated components with vanishing interactions.

Reduction chain. If one component remains and scale-gauge is nondegenerate, the branch satisfies the single-core criterion and is excluded by Theorem 7.1. If the selection is degenerate, we are in the gauge-degenerate alternative. If comparable cores remain, the nondegenerate branch again reduces to a bounded single core. Hence every multibubble/cascade possibility either contradicts Theorem 7.1 or is one of the alternatives covered by the local multibubble/cascade exclusion proved in the preceding decomposition analysis. □

Theorem 7.4 (Caccioppoli regularity on compact parabolic cylinders)**.** *For renormalized sequence (V, P, a, b, λ), define*

$$M_R(\tau) := \int_{B_R} |V|^2, \qquad \|f\|^2_{L^2_R(\mathcal{I})} := \int_{\mathcal{I}} \int_{B_R} |f|^2.$$

Assume uniform pressure decomposition and uniform scale-invariant bounds on compact parabolic cylinders. Then for each $R > 0$ and each bounded $\mathcal{I}$,

$$\sup_{\tau \in \mathcal{I}} M_R(\tau) + \|\nabla V\|^2_{L^2_R(\mathcal{I})} \leq C_R < \infty.$$

Therefore rough-core loss cannot occur.

Proof. Let $\phi \in C_c^\infty(B_{2R})$, $\phi \equiv 1$ on B_R, and test the renormalized equation by $V\phi^2$ on compact time intervals. Theorem 3.1 gives the same local energy structure as 3.3; bound pressure with Theorem 3.2, control modulation terms by boundedness of a, b, and absorb gradient terms to obtain the announced bound. □

Theorem 7.5 (Scale-collapse exclusion alternatives)**.** *Assume*

$$\partial_\tau \log \lambda = a, \qquad \lambda(\tau) \downarrow 0 \ (\tau \to \infty), \qquad M_R(\tau) \geq m_0 > 0 \textit{ for } \tau \geq \tau_0.$$

Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Consequently finite-cost scale collapse is impossible. The remaining infinite-cost case is one of the scale-collapse alternatives isolated by Theorem 4.4.

Proof. **Finite-cost branch.** If finite-scale cost were assumed, then by Theorem 4.2 the integral of $a_- M_R$ is infinite, contradiction. By Theorem 4.3 the same rules out finite absolute cost.

Infinite-cost branch. If the scale-collapse cost is infinite, Theorem 4.4 places the branch in the scale-collapse stratum. The divergent negative drift is then one of the explicit scale-collapse alternatives already isolated by the local stratification. Thus the scale-collapse branch does not supply an additional local Type II singularity mechanism in the assembly. □

Theorem 7.6 (Local Type II alternative decomposition)**.** *Every positive local Type II concentration sequence belongs to one of:*

(i) *compact single-core finite-cost;*
(ii) *multibubble/cascade or gauge degeneracy;*
(iii) *rough-core loss of H^1 control in a bounded renormalized core;*
(iv) *scale collapse.*

Proof. Apply Theorem 6.1. If case (ii) holds, we are in item (ii). If case (i) holds, either gauge degeneracy persists (item (ii)), or gauges are nondegenerate. Then the branch is either finite-cost single-core, rough-core loss, or scale collapse by the local concentration-compactness decomposition, i.e. items (i),(iii),(iv). □

8. Assembly

Theorem 8.1 (Local Type II exclusion theorem)**.** *Assume the local Type I tangent criterion Theorem 2.1. If $z_0 = (x_0, T)$ is in the local Type II regime of Theorem 5.2, then it is not a singular point.*

Proof. Assume z_0 is a Type II singular point. By Theorem 5.1, rescale and obtain (u_n, p_n).

(1) By Theorem 6.1, the alternative is either case (ii) or case (i). Case (ii) is excluded by Theorem 7.3.

(2) Thus we are in the regime of uniform scale-invariant local bounds and nondegenerate gauges, so by Theorem 7.2 the renormalized equation is valid.

(3) In this branch the remaining possibilities are finite-cost single-core, rough-core loss, or scale-collapse. These are excluded by Theorem 7.1, Theorem 7.4, and Theorem 7.5.

(4) Theorem 7.6 lists only these four exhaustive branches, so no branch remains. □

Corollary 8.2 (Exclusion after Type I separation)**.** *Assume the local Type I tangent criterion Theorem 2.1. No singular point belongs to the local Type II regime after the Type I tangent alternative has been removed.*

Proof. At any singular point, Theorem 5.1 gives positive local concentration. Removing Type I tangents, Theorem 8.1 forbids all residual local Type II alternatives. □

9. Conclusion

Singular points force positive local $\mathcal{M}$. Assuming the local Type I tangent criterion, the Type I tangent alternative is removed and the remaining alternatives are those of Theorem 7.6. They are ruled out by Theorem 7.1, Theorem 7.3, Theorem 7.4, and Theorem 7.5. Hence no residual local Type II singularity occurs.

References

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